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Abstract

Among additive manufacturing (AM), 3D inkjet technology, materials extrusion (ME), and digital light processing (DLP), which are from dot and line to face printing, have been extensively investigated for biological and pharmaceutical applications. These techniques are valued for their ability to create customised, complex, drug-laden devices and tissue engineering scaffolds. However, testing new bioinks or filament designs can be both expensive and time-consuming. To this end, numerical simulation offers a useful solution by reducing costs and saving time. Both machine learning (ML) and theory-based models can be used for simulation. ML excels in handling complex data but faces challenges with data availability and overfitting, while theory-based models provide a more interpretable and data-efficient framework. This review explores how theory-based numerical simulation can be used to assess and optimise factors such as bioink printability, technique mechanism, printing parameters, and post-printing outcomes. By using simulation, key parameters can be understood and optimised without performing extensive physical experiments. The review highlights current models and discusses opportunities and challenges in using simulations to enhance the AM process, potentially advancing regenerative medicine and personalised treatments.

1. Introduction

Additive manufacturing (AM) is a rapidly developing technology that revolutionises product design and the manufacturing process. The production of AM products typically involves four steps [1]. First, computer-aided design (CAD) software is used to create the digital model, which serves as the input for the 3D printer. Second, a suitable material is chosen to meet the desired requirements of the product, with options ranging from inks and filaments to powders. Third, an appropriate printing method is selected. According to ISO/ASTM 52 900:2015, seven methods are defined: a. binder jetting (BJ); b. directed energy deposition (DED); c. material extrusion (ME); d. material jetting (MJ); e. powder bed fusion (PBF); f. sheet lamination (SL); and g. vat photopolymerisation (VP). Finally, the materials are printed layer by layer to construct the 3D structure [2, 3].

AM offers several advantages over conventional manufacturing, in terms of rapid prototyping, customisation and material conservation [4, 5]. First, AM streamlines the manufacturing process by directly converting a virtual model into a physical product, enabling the quick realisation of novel concepts. Next, AM allows tailored designs without the need for additional tooling. Finally, AM minimises material usage. Consequently, AM has been extensively researched for producing functional parts using a variety of materials, such as high-temperature alloys, biomaterials, ceramics, and concrete [5, 6].

However, the full potential of AM technology is not yet completely understood, particularly in terms of the relationship between building materials and final products.

1.1. Simulation overview

Simulation of temporal dynamics of real-world processes and systems using mathematical models is widely applied across various fields, including aerospace, automotive, energy, manufacturing, and biomedical engineering. Simulation is important and is particularly useful when performing a direct real-world experiment or assessment is infeasible, costly, or unsafe. Simulation studies are performed for different purposes such as providing insights about system dynamics, assessing performance of a model, predicting outcomes of an experiment, testing an algorithm, or modelling a natural system. By developing a suitable model, it is possible to predict creep behaviour, mechanical strength, fluid dynamics, cell seeding behaviour / identification, and optimise structures of targeted products [7–15].

Simulation approaches can be categorised according to their underlying modelling philosophy, typically, theory-based, data-driven, and hybrid methods. Theory-based simulations, also known as physics-based or principle-based models, rely directly on the fundamental laws of nature, such as fluid dynamics, thermodynamics, and continuum mechanics. These models can be further subdivided into analytical approaches, which yield closed-form solutions, and numerical approaches, which obtain approximate solutions through computation [16]. An analytical method solves a problem exactly in closed-form equations, offering clear physical insight but typically requiring simplifying assumptions and simple geometries. A numerical method approximates the solution using computational algorithms, allowing complex systems to be handled but yielding results as discrete numerical values rather than formulas. In practice, analytical methods are often used for insight and validation, while numerical methods are used to solve complex problems.

In contrast, data-driven, or empirical-based simulations use statistical models, machine learning (ML) techniques, and empirical relationships to learn patterns from experimental measurements or high-fidelity simulation outputs. Empirical-based models describe system behaviour by learning relationships directly from experimental or observed data, rather than deriving them from first principles or physical laws. They are often implemented using statistical regression or machine-learning techniques and are useful when the underlying physical mechanisms are complex, or impractical to model analytically [16]. Bridging these two worlds, hybrid simulations integrate theory-based formulations with data-centric components, enabling improved accuracy, efficiency, and scalability, an approach increasingly common in multiscale modelling and digital twin systems.

Within the scope of bioprinting, *in-silico* modelling of 3D printing processes has attracted significant attention in biomedical AM [6, 17, 18]. Conducting numerous repetitions with varying parameters in AM fabrication can be both time-consuming and expensive. Simulations can assist in optimisation of AM materials and fabrication techniques for various biomedical applications, such as materials design, custom implants, drug delivery, and tissue engineering. These optimisations accelerate the development of new bioinks, construct *in vitro* models, enable personalised treatments, increase therapeutic efficacy, and enhance patient outcomes.

1.2. ML

As a data-driven method, ML is a valuable simulation tool that can predict specific outcomes in 3D bioprinting by processing large datasets through different algorithmic models. This approach allows for the identification of complex patterns and relationships within large amounts of data, making it possible to anticipate results in scenarios involving biomaterials, tissue growth, and structural stability. Previous researchers have intensively explored ML for optimisation of AM and prediction of AM outcomes [6, 19–25]. However, ML relies heavily on extensive training datasets, which are often difficult or impossible to obtain, especially in some complex, challenging, expensive, or time-consuming conditions where experimental data is still limited. These models can also be prone to overfitting, where predictions are too specific to the training data and may fail to generalise well to new scenarios.

1.3. Theory-based simulations

On the other hand, conventional theory-based models use well-established scientific principles and equations, which makes them more interpretable and reliable. Since they are derived from foundational physical, chemical, or biological laws, these models often allow researchers to understand how specific parameters directly affect outcomes. This transparency can be especially valuable in biomedical fields, where regulatory requirements emphasise predictability and consistency [26]. Moreover, traditional simulations generally require far less data than ML models. They can produce meaningful predictions even with limited datasets, achieving reasonable accuracy by leveraging prior theoretical knowledge. These conventional approaches, based on validated models, often offer a more interpretable framework, allowing researchers to rely on prior knowledge for more consistent predictive accuracy in specific bioprinting contexts and explain the relationship between input parameters and the results. Furthermore, validated models are generally generalisable, as they can be adapted to different conditions by adjusting parameters and boundary conditions while maintaining the theoretical framework. The scalability also allows proposed models to be applied across various scales and levels of complexity, making them a versatile tool in scientific research and practical applications.

1.4. Modelling inkjet, extrusion, and DLP processes

While there are seven commonly recognised AM methods, PBF, BJ, DED, and SL are primarily used for processing metals, ceramics and composites, which are not commonly used for handling cells, hydrogels, or other biologically relevant materials [2, 3]. Biomedical and pharmaceutical printing primarily utilises three key techniques, namely, inkjet printing, ME, and VP, which includes digital light processing (DLP) and stereolithography. These methods are the top three adopted in this area due to their compatibility with soft, cell-laden biomaterials, ease of access and operation, lower cost relative to other bioprinting methods, and ability to fabricate complex tissue constructs with high precision [27, 28].

In this review, we first introduce the theories that are generally applied in the bioprinting field. Then, we explore the applications and optimisation of theory-based simulations in the biomedical area, focusing on 3D inkjet technology, ME, and DLP printing (stereolithography has similar photopolymerisation mechanism as DLP while laser assisted bioprinting has relatively limited availability of comprehensive simulation models). Each AM method is analysed across pre-processing, in-process, and post-printing stages as shown in figure 1 [29–31]. The key controlling parameters, along with their interrelationships and governing equations, were described. Finally, we extend our expertise and insights across different printing methods to a comparative analysis of how similar modelling strategies are employed differently based on the printing mechanism, material properties, and biological limitations. We aim to provide information that can assist both newcomers and experienced users in improving their bioprinting processes. Elucidating the fundamental scientific principles and governing equations of bioprinting is essential for transitioning the field from empirical trial-and-error towards highly predictive and controlled manufacturing methodologies.

To provide a wider perspective, we introduce a classification based on simulation strategy. By categorising approaches according to the physical state of the simulated objects, namely, liquid, solid, particle, and multiphase systems, we can better represent the underlying physical behaviours being modelled. This state-based taxonomy serves as a vital complement to traditional AM classifications organised by printing technology. For instance, in liquid-based models, we use frameworks derived from the Navier–Stokes equations to characterise bioink flow and droplet dynamics. In solid-based models, we rely on continuum mechanics to evaluate structural deformation and long-term mechanical stability. The specific technical distinctions and methodologies for each category are detailed in table 1.

2. Theories used in the 3D bioprinting simulation

In this section, we will review the concepts, theories, and simulation models that are most used to analyse and optimise above-mentioned 3D bioprinting processes. The fundamental theories we considered here are widely adopted in numerical simulation studies, particularly those that address key aspects of the printing process such as material flow, droplet dynamics, structural integrity, and drug release kinetics. The inclusion of each model or concept was further guided by its suitability for capturing physical phenomena relevant to bioink behaviour, processing parameters, and final product performance. To further clarify the theoretical foundations behind these stages, table 1 links each stage of the printing process in figure 1 with the corresponding theoretical models and governing equations.

2.1. Dimensionless numbers

Dimensionless numbers are a ratio of physical quantities providing estimates of systems' behaviour without units. They can help in predicting the consistency of droplet formation and extrusion from the nozzle in inkjet printing. In the context of bioink printing, Reynolds (Re), Weber (We), and Ohnesorge (Oh) numbers are used to characterise the behaviour of the bioink during the printing process [32, 33]. These numbers are defined as below:

$$Re = \frac{\rho R V}{\eta} \tag{1}$$

$$We = \frac{\rho R V^2}{\sigma} \tag{2}$$

$$Oh = \frac{\sqrt{We}}{Re} = \frac{\eta}{\sqrt{\sigma \rho R}} \tag{3}$$

where, ρ is the liquid density, R is characteristic length, V is the liquid velocity, η is the liquid dynamic viscosity, and σ is the surface tension of the bioink. The Reynolds number represents the ratio of inertial forces to viscous forces in the fluid. It is used to assess the flow regime, helping to determine whether the flow is laminar or turbulent, which influences the stability of droplet formation. The Weber number is the ratio of inertial forces to surface tension forces. It helps in predicting the tendency of the droplet to break up during extrusion, with higher We numbers indicating a greater likelihood of droplet disintegration. The Ohnesorge number combines the effects of

Table 1. Key governing equations and modelling principles align with pre-processing, in-process, and post-printing stages as shown in figure 1.

Printing method	Process stage		Key physical phenomenon	Typical governing equations/models	Applicability	References
Inkjet printing	Pre-processing	• Bioink printability	Rheological behaviour and jet stability	Rheological models, dimensionless analysis (*Re, We, Oh*) $$\mathrm{Re}=\frac{\rho RV}{\eta},\ \mathrm{We}=\frac{\rho RV^2}{\sigma},\ \mathrm{Oh}=\frac{\sqrt{\mathrm{We}}}{Re}=\frac{\eta}{\sqrt{\sigma\rho R}}$$	Liquid	[32, 33]
	In-process	• Effect of printing parameters • Flow behaviour of inks • Principle of ink printing	Pressure, velocity, nozzle dynamics Non-Newtonian fluid behaviour Droplet generation mechanism	Navier–Stokes equations, fluid momentum equations $$\rho\frac{Du}{Dt}=\rho F_v-\nabla p+\mu\nabla^2 u$$ $$f_i\left(x+\triangle t\vec{c}_i,t+\triangle t\right)=f_i\left(x,t\right)+\Omega_i$$	Liquid /Particle	[34–37]
	Post-printing	• Product behaviour	Structural integrity and functionality	Mechanical stability models $$\sigma=\frac{F}{A},\ \tau=\frac{F}{A},\ \varepsilon=\frac{\triangle L}{L}$$ $$\sigma=E\varepsilon,\ \tau=G\gamma,\ G=\frac{E}{2\left(1+\upsilon\right)}$$	Solid	[38, 39]
Material extrusion	Pre-processing	• Materials printability	Viscosity and shear-thinning behaviour	Rheological models $$\mathrm{Re}=\frac{\rho RV}{\eta},\ \mathrm{We}=\frac{\rho RV^2}{\sigma},\ \mathrm{Oh}=\frac{\sqrt{\mathrm{We}}}{Re}=\frac{\eta}{\sqrt{\sigma\rho R}}$$	Liquid	[32, 33]
		• Drug loading contents	Mass distribution in filament	Diffusion and transport models $$\frac{\partial\varphi}{\partial t}=D\Delta\varphi$$	Multi-phase /Solid	[40, 41]
	In-process	• Flow behaviour of materials	Extrusion flow through nozzle	Navier–Stokes equations, non-Newtonian flow models $$\rho\frac{Du}{Dt}=\rho F_v-\nabla p+\mu\nabla^2 u$$ $$f_i\left(x+\triangle t\vec{c}_i,t+\triangle t\right)=f_i\left(x,t\right)+\Omega_i$$	Liquid /Particle	[34–37]

(Continued.)

Table 1. (Continued.)

	Post-printing	• Thermo and mechanical properties • Printing quality of final products	Temperature and structural stability Shape fidelity and filament stability	Heat transfer models, Continuum Mechanics $\rho C_p\left(\frac{\partial T}{\partial t}\right) = \nabla\left(k\nabla T\right) + h\nabla T + Q$ $\sigma = \frac{F}{A},\ \tau = \frac{F}{A},\ \varepsilon = \frac{\triangle L}{L}$ $\sigma = E\varepsilon,\ \tau = G\gamma,\ G = \frac{E}{2\left(1+\upsilon\right)}$	Solid /Liquid	[38, 39, 42]
		• Drug release profile	Drug diffusion and degradation	Diffusion models $\frac{\partial\varphi}{\partial t} = D\Delta\varphi$	Multi-phase /Solid	[40, 41]
DLP	In-process	• Bioink photopolymerisation	Light-induced curing reaction	Photopolymerisation kinetics, Beer–Lambert law $v_p = k_p\left(\frac{\phi\varepsilon I_0 C_i}{N_A h v k_t}\right)^{1/2}[M]^{3/2}$ $\log_{10}\left(I_0/I\right) = A = \varepsilon lc$	Liquid /Particle	[43–45]
		• Flow behaviour of bioink	Resin viscosity and layer formation	Navier–Stokes equations, fluid momentum equations $\rho\frac{Du}{Dt} = \rho F_v - \nabla p + \mu\nabla^2 u$ $f_i\left(x + \triangle t\vec{c}_i, t + \triangle t\right) = f_i\left(x,t\right) + \Omega_i$	Liquid /Particle	[34–37]
	Post-printing	• Shape deformation	Shrinkage and curing stress	Solid mechanics models $\sigma = \frac{F}{A},\ \tau = \frac{F}{A},\ \varepsilon = \frac{\triangle L}{L}$ $\sigma = E\varepsilon,\ \tau = G\gamma,\ G = \frac{E}{2\left(1+\upsilon\right)}$	Solid	[38, 39]
		• Drug release profile	Drug diffusion from cured matrix	Diffusion and degradation models $\frac{\partial\varphi}{\partial t} = D\Delta\varphi$	Multi-Phase /Solid	[40, 41]

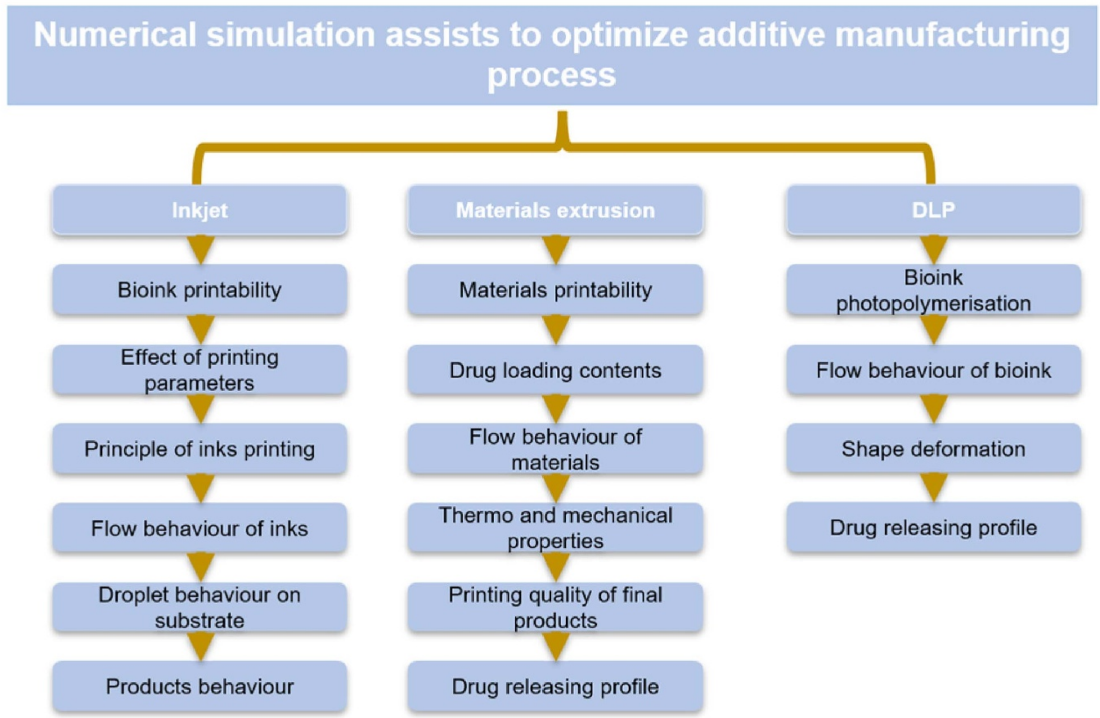

Figure 1. The flow chart outlining contents of the review, covering 3D inkjet printing, ME, and DLP printing processes [29–31].

viscosity, surface tension, and inertial forces, providing a dimensionless measure of the relative importance of viscous forces in droplet formation and stability. A low *Oh* number typically indicates that the fluid can form droplets easily, while a high *Oh* number suggests that viscous forces may dominate, potentially inhibiting droplet formation [32, 33].

2.2. Temperature evolution

3D printing processes often involve phase changes, with heat being a primary factor driving these transformations. When a material is heated above melting temperature, it undergoes a phase change from solid to liquid. Conversely, when the temperature decreases below the melting point, the material transitions from a liquid to a solid phase. In 3D bioprinting, accurately modelling the time-dependent temperature profile is crucial for predicting these phase changes. It is typically based on Fourier's second law of heat conduction [42]:

$$\rho C_p \left(\frac{\partial T}{\partial t}\right) = \nabla (k \nabla T) + h \nabla T + Q \qquad (4)$$

where ρ is the density, C_p is the heat capacity, T is the temperature, t is the time, k is the thermal conductivity coefficient, h is the convection coefficient, Q is the internal heat source. Fourier's second law provides a mathematical framework for understanding how heat diffuses through a material over time, which is essential for controlling the phase transitions that occur during the bioprinting process.

2.3. Fluid dynamics during printing

Considering the flow of the bioink or melted filament in the nozzle, it is generally regarded as incompressible and exhibiting non-Newtonian fluid behaviour. The flow behaviour can be effectively modelled using the Navier–Stokes equation with modifications to the fluid viscosity coefficient to account for the non-Newtonian characteristics [34, 35]. The Navier–Stokes equation describes the conservation of momentum and mass for fluids by:

$$\nabla u = 0 \qquad (5)$$

$$\rho \frac{Du}{Dt} = \rho F_v - \nabla p + \mu \nabla^2 u \qquad (6)$$

where, u is the velocity of the fluid, ρ is the flow density, $\frac{D}{Dt}$ is the material derivative, defined as $\frac{\partial}{\partial t} + u \cdot \nabla$, F_v is the external forces, p is the pressure, μ is fluid viscosity coefficient. For non-Newtonian fluid, μ depends on the rate of fluid deformation at specific temperature. Equation (5) ensures the conservation of mass for an incompressible fluid, requiring that the divergence of the velocity field is zero. Equation (6) represents the conservation of momentum, capturing the balance between inertial forces, pressure gradients, viscous forces, and any external forces. These equations are essential for accurately modelling the flow of bioinks and melted filaments during the 3D bioprinting process, where understanding the fluid dynamics is crucial for optimising printing performance and material behaviour.

2.4. Fick's second law

In bioprinting, understanding the diffusion behaviour is essential for predicting the distribution of nutrients, drugs, or other molecules within printed tissues and ensuring uniformity and efficacy in the final constructs. Fick's second law describes the time-dependent passive diffusion of small molecules from regions of high concentration to low concentration. This law can be derived from Fick's first law, as shown in equation (7), which describes the diffusion flux, and the principle of mass conservation, assuming no chemical reactions are involved [40, 41]:

$$\frac{\partial \varphi}{\partial t} = D\Delta\varphi \tag{7}$$

where, φ is the concentration that depends on location and time, t is time, D is the diffusion coefficient. The Fick's second law has similar mathematical form as heat equation, where we replace the thermal conductivity k with diffusion coefficient D. Equation (7) describes how the concentration of a substance changes with time due to diffusion, providing a critical framework for modelling diffusion-driven processes in various applications.

2.5. Mechanics and strength of materials

The printed structures or products should meet the requirement of the mechanical properties to withstand various loading conditions. In continuum mechanics, materials subjected to external forces undergo deformation, which can be quantitatively described by stress, strain, and shear stress, as shown in equations (8)–(10), respectively [38].

$$\sigma = \frac{F}{A} \tag{8}$$

$$\tau = \frac{F}{A} \tag{9}$$

$$\varepsilon = \frac{\triangle L}{L} \tag{10}$$

where, σ is the stress, τ is the shear stress, ε is the strain, F is the applied force, A is the applied cross-section area, $\triangle L$ is the deformation, is the original length. These quantities are fundamental in assessing how a material deforms and behaves under various forces, ensuring that printed structures achieve the necessary mechanical integrity and performance for their intended applications. When a solid material is subjected to a small load and exhibits elastic deformation, the relationship between stress and strain is typically linear and can be expressed as [39]:

$$\sigma = E\varepsilon \tag{11}$$

$$\tau = G\gamma \tag{12}$$

where, is Young's modulus, which exhibits the ability to resist the deformation of applied force. G is shear modulus, and γ is shear strain, which is defined by the change of the angle. In this linear elastic regime, the material will return to its original shape once the load is removed, provided the applied stress does not exceed the material's elastic limit. The relationship between E and G is given by equation (13), where υ is Poisson's ratio [38, 39].

$$G = \frac{E}{2(1+\upsilon)}. \tag{13}$$

2.6. Finite element methods (FEM)

FEM is a widely used mathematical simulation method in engineering area, particularly for solving complex structural or time-dependent problems that do not conform to simple geometries and cannot be addressed through closed-form solutions [46]. FEM is highly regarded in the engineering community due to its versatility and effectiveness in handling a wide range of circumstances and problems. With advancements in high-performance computing, FEM has become an integral component of modern CAD and computer-aided manufacturing (CAM).

Before applying FEM, it is essential to define the variables, underlying theory, solution domain, and boundary conditions. The application of FEM typically involves three key steps [47, 48]. The first step is domain discretisation. The structure or continuous body of interest is divided into several subdomains or elements. These elements are interconnected at shared edges and nodes, forming a mesh that represents the entire problem domain. The second step is to apply the factors. Essential parameters, including material properties, boundary conditions, and environmental factors, are applied to the meshed structure. These inputs are critical for accurately modelling the physical behaviour of the system under study. The last step is to solve the model. The model is solved by applying the equilibrium equations to the discretised domain. This analysis allows for the calculation of the desired quantities, such as stress, strain, temperature distribution, or displacement, providing insights into the behaviour of the system under the specified conditions.

2.7. Lattice Boltzmann method (LBM)

Modelling fluid flow using the Navier–Stokes equations in structures with porous surfaces, complex boundaries, or under multiphase conditions poses significant challenges. These scenarios often involve intricate geometries and interactions that are difficult to capture using traditional methods. The LBM provides an alternative approach for simulating fluid dynamics in such complex situations [49–51]. LBM simulates fluid behaviour on a discrete lattice, where fluid particles move and interact in a manner that bridges the gap between macroscopic continuum mechanics and microscopic molecular dynamics. Like FEM, the geometry of the domain

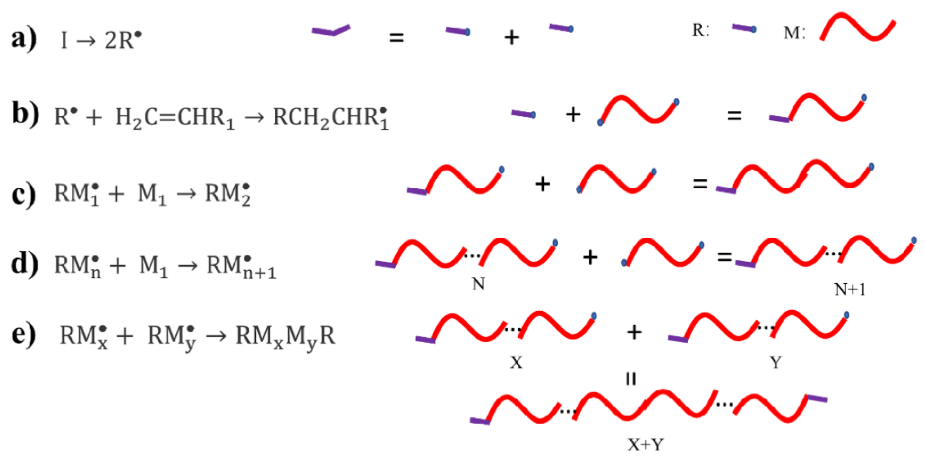

Figure 2. Schematic of the chain-growth polymerisation mechanism. (a) Photoinitiator molecular decomposes to two free radicals. (b) Free radicals initiate polymer chain growth. (c) and (d) the radicalised monomer link to nearby monomers. (e) The termination reaction of polymer chain growth. (*R*: free radicals, *M*: monomer).

must be discretised into elements or lattices. Each node in the lattice represents a specific point in space where fluid particles can reside. In LBM, fluid particles move from one node to another along predefined lattice links, simulating the fluid's advection. When particles meet at a node, they undergo a redistribution of velocities according to a collision rule, which is designed to conserve mass and momentum. This collision step effectively mimics the advection and collision processes of particles, allowing LBM to capture complex fluid behaviours in porous media, irregular boundaries, and multiphase flows with high accuracy. The general formula of the Lattice Boltzmann equation is:

$$f_i\left(x+\triangle t\overrightarrow{c}_i, t+\triangle t\right) = f_i\left(x,t\right) + \Omega_i \tag{14}$$

where f_i is the concentration of particles that travels with velocity $\overrightarrow{c}_i$ at the x poistion at time t, $\triangle t$ is the time step, Ω_i is the collision operator. The LBM's ability to handle complex boundaries and porous structures, along with its inherent parallelism, makes it a powerful tool for simulating fluid dynamics in scenarios where traditional methods like the Navier–Stokes equations fails.

2.8. Free radical chain growth polymerisation

Most photo-crosslinking materials systems are based on free radical chain growth polymerisation theory to form hydrogels. In specific, the photoinitiators will decompose into free radicals under the specific wavelength of light exposure (e.g. I2959 at 365 nm, LAP at 365 and 405 nm) [52]. Then, these free radical initiates the polymerisation process by linking monomers to form polymer chains.

Chain growth polymerisation process can be divided into 3 stages [53]: 1. initiation; 2. chain growth; and 3. termination (figure 2). In the initial stage, the photoinitiator decomposes into two free radicals under specific wavelengths of light (figure 2(a)). Then, the free radicals react with and rearrange monomers, which typically have the carbon–carbon double bonds structures (figure 2(b)). The radicalised monomer links to nearby monomers to form a chain like structure (figures 2(c) and (d)). To terminate the chain growth polymerisation process, the radicalised monomers need to meet with: 1) another radicalised monomer (figure 2(e)); 2) free radicals; 3) a monomer or polymer chain for free radical transfer; or 4) impurities and inhibitors (e.g. oxygen). Besides, the termination can also occur via hydrogen abstraction, which forms two separate terminated polymer chains. The mode of termination, whether by monomer combination or hydrogen abstraction, depends on the molecular type and reaction temperature [43].

The kinetics of a photoinitiator can be described by the following equation [43]:

$$R_i = \frac{2\phi\varepsilon I_0 f C_i}{N_A h\nu} \tag{15}$$

where, R_i is the initiation rate, ϕ is quantum yield, ε is extinction coefficient, I_0 is the light exposure intensity, f is the photoinitiator efficiency, C_i is the concentration of the photoinitiator, N_A is the Avogadro's number, h is the Planck's constant, and ν is the frequency of initiating light. Based on the steady-state assumption, the relationship between the initiation rate and the polymerisation rate (ν_p) can be determined by:

$$\nu_p = k_p\left[M\right]\left(\frac{R_i}{2k_t}\right)^{1/2} \tag{16}$$

where, k_t is the termination rate coefficient, and M is the monomer concentration. For a low efficiency initiator f becomes a function of $[M]$. Substituting equation (15) into the equation (16), the rate of photopolymerisation can be determined by the following equation [43]:

$$v_p = k_p \left(\frac{\phi \varepsilon I_0 C_i}{N_A h v k_t} \right)^{1/2} [M]^{3/2}. \qquad (17)$$

From the equation (17), the photopolymerisation rate is proportional to the square of the light exposure intensity and the concentration of the photoinitiator. In addition, the rate is also dependent on the concentration of photocrosslinkable monomer to the power of 1.5. It indicates that increasing the concentration of photocrosslinkable monomer, light exposure intensity, concentration of the photoinitiator, and selecting appropriate materials can enhance the efficiency of the photo-crosslinking process.

2.9. Beer–Lambert law

Light based 3D bioprinting relies on adjusting light intensity, exposure time, and light absorption rate to control the printing efficiency and accuracy. The Beer Lambert law describes the absorption of light by a chemical species in a solution as a function of its concentration. This law is crucial in various fields such as chemistry, physics, and biology for analysing solutions and determining the concentrations of substances. The expression of Beer–Lambert law is [44, 45]:

$$\log_{10}(I_0/I) = A = \varepsilon l c \qquad (18)$$

where, I_0 is the intensity of the incident light, I is the intensity of the transmitted light, A is the absorbance, ε is the molar attenuation coefficient or absorptivity of the attenuating species, l is the optical path length, and c is the concentration of the attenuating species. Based on the Beer–Lambert's Law, the light energy used for photopolymerising an infinitesimal layer can be expressed by considering the absorption of light as it passes through the photopolymerisable material. The energy absorbed by a thin layer of thickness dh at depth h can be expressed as [54]:

$$\mathrm{d}I = -\varepsilon \cdot c \cdot I(h) \cdot \mathrm{d}h \qquad (19)$$

where dI is the differential decrease in light intensity as it passes through an infinitesimal layer of thickness, and $I(h)$ is the light intensity at depth h within the material. Integrating this expression over the entire thickness of the material would give the total energy absorbed, but for an infinitesimal layer, this differential form captures the energy required to initiate photopolymerisation at that specific depth. The total energy absorbed in a layer of thickness dh can be used to model the rate of photopolymerisation and to optimise the exposure parameters in processes that rely on precise light-based curing techniques [54].

2.10. Monte Carlo method

Monte Carlo methods use repeated random sampling to obtain numerical estimate of parameters and approximate solutions for solving problems that direct calculation of their solutions is difficult or impossible [55]. For example, Monte Carlo simulations can be applied to calculate the formation and distribution of crystal cores in printed plastic crystal materials, where the stochastic nature of nucleation and growth processes introduces uncertainty [56]. Similarly, in bioprinting, Monte Carlo methods can model the distribution of printed cells within a hydrogel matrix, accounting for the randomness in cell placement and movement during the printing process. Additionally, in DLP printing, Monte Carlo simulations can help assess the influence of cell membranes on light distribution, which is critical for accurately predicting the curing patterns in the photopolymerisation process [57]. By implementing these methods through computer programming, approximate solutions to these complex problems can be achieved.

3. The application of simulation methods for 3D bioprinting

The operational process of bioink-based 3D printing devices can generally be divided into three major steps, namely, preparation, in-process, and post-processing. In the preparation phase, a detailed analysis of the bioink's properties is conducted to assess its suitability for the printing process. This involves modelling the rheological behaviour of the bioink, which is crucial for maintaining uniformity in fluid dynamics and preserving cell viability. The in-process phase focuses on modelling the formation, morphology, and movement of droplets/bioinks, which are key factors that influence the precision, resolution, and overall quality of the printed microstructures. Simulation of various printing parameters during this stage plays a critical role in optimising the process by enhancing print speed, minimising errors, and reducing operational costs. In the post-processing phase, the properties of cell- or drug-laden bioinks can alter post-extrusion, especially when exposed to crosslinking agents or changes in external conditions. Modelling the relevant physicochemical parameters at this stage is essential for optimising drug release profiles, ensuring cell viability, and enhancing the mechanical properties of the final printed constructs. Each subsequent parallel section on inkjet printing, ME, and DLP connects these above-mentioned theories to current simulation studies in the respective techniques. This approach helps illustrate how each model is applied in practice and highlights howeach simulation model assists in optimising 3D bioprinting process.

3.1. 3D inkjet technology

Inkjet bioprinting is a non-contact, droplet-based technique widely used for depositing bioinks in a controlled manner. It operates by ejecting small droplets of bioink through a nozzle onto a substrate. This method allows precise spatial control of bioink deposition, making it suitable for fabricating complex biological structures.

The printing process in inkjet bioprinting begins with the design and preparation of the bioink. Unlike traditional biomaterials, bioink may contain cells and operates under varying temperature, pressure, and flow speed conditions. To assess whether a given bioink formulation is printable under a specific condition, key parameters such as nozzle geometry and bioink properties (e.g. viscosity, density, surface tension) are used to calculate dimensionless numbers in equations (1)–(3). These values can then be used to generate a printability map, such as the diagram shown in figure 3(a) [58, 59]. Four dashed lines divide the diagram into five distinct areas based on droplet formation characteristics: The centre grey area represents the standard and desirable printable bioink region where stable, satellite-free droplets are formed. In contrast, the surrounding areas represent regions of suboptimal or failed droplet formation: (1) low velocity: low *Re* values result in inadequate momentum to form stable droplets; (2) splashing on substrate: at high *Re* values, excessive kinetic energy causes droplets to splash upon impact; (3) too viscous: high *Oh* values indicate excessive viscosity, preventing proper droplet formation where droplets breakup cannot occur and (4) Satellite droplets: at low *Oh* and moderate-to-high *Re*, droplet breakup leads to undesirable satellite formation. The overlapping areas between regions indicate cases where multiple printability issues coexist, such as both splashing and satellite formation. However, the boundary lines in this diagram are not absolute but depend on specific bioink properties. Besides, the viscosity of the bioink plays a key role in determining the size and location of the stable printing zone, with higher viscosity typically reducing the printable area [58].

Another study assessed the relationship between formulation printability with its dynamic viscosity, surface tension, ink density, and nozzle diameter under various pressure and temperatures via a proposed high throughput system [61]. When bioink meets with various nanoparticles, it is significant to consider the influence of the nanoparticles on the viscoelasticity, which later affects the printability of bioinks. Particle volume fraction, particle size, and its distribution were included in the model to make predictions [62]. Utilising the proposed model can greatly accelerate the development of new bioink formulations, with or without additives, and results in cost reduction.

After examining the printability of bioink, and loading the bioink into the printer, the 3D inkjet printing is ready to work. During the printing process, the droplet morphology is important as it forms the printed structures and determined printing resolution, surface quality, and structural integrity of the final products. The key parameters in the printing process, including pressure, viscosity of bioink, temperature, and nozzle morphology, can determine the printability, cell viability and shape fidelity.

The viscosity of bioink and inlet pressure are key factors affecting the printing speed [63]. Increasing the viscosity of the bioinks slows down the droplet formation, whereas increasing surface tension accelerates it [64]. High-viscosity bioink requires high inlet pressure to maintain a consistent extrusion rate, which in turn increases shear stress through the printing path due to the increased fluid speed. The high shear stress can influence the distribution of particles within the bioink. Increasing the inlet temperature enhances extrusion velocity, reduces shear stress, and decreases outlet pressure [65, 66].

A nomogram has been developed to illustrate the relationship between 1. outlet diameter; 2. pressure difference between inlet and outlet; 3. mass flow rate; and 4. maximum shear stress based on the Herschel–Bulkley model (figure 3(b)) [60]. This model can be adjusted to create new nomograms for different bioinks.

Typical droplet formation in inkjet printing involves three steps: 1. negative pressure forms a meniscus surface; 2. positive pressure creates a jet window; and 3. negative pressure breaks the droplet and forms a new meniscus surface for next droplet. This dynamic phenomenon can be expressed by the Navier–Stokes equation or LBM and exhibited through simulations (figure 4(a)) [36, 37]. Due to the velocity difference between the centre to the edge of the fluid, shear force thins the stream until it reaches a critical value, where the surface tension pinches it into droplets. In laser-induced inkjet printing, the high-energy laser strikes the liquid layer, creating a bubble that forces the bioink downward until the bubble bursts. The remaining momentum causes the bioink to continue moving downward, forming a droplet when the momentum exceeds the restrictive forces of viscosity and surface tension (figure 4(b)). Simulating this process can assist in optimising the energy input, droplet formation, and bioink layer thickness to control droplet size accurately [67]. To minimise the thermal effect of the laser on the ink, a dual chamber system with propellant liquid and bioink separated by a plastic membrane was designed (figure 4(c)) [68]. When the propellant liquid is exposed to the laser, a phase change from liquid to gas occurs, creating a pressure variation that extrudes bioink through the nozzle. The study also examined how nozzle diameter, membrane thickness, and laser energy affect the printing process, finding that a smaller nozzle diameter requires less laser energy, thus minimising thermal impact.

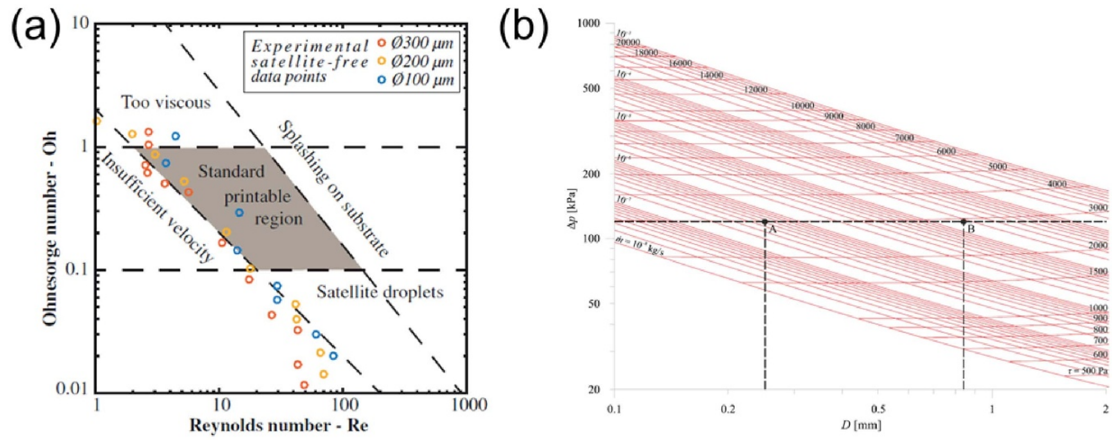

Figure 3. (a) Illustration of the standard inkjet-printability diagram which can be divided into 5 distinct areas. The grey area in the middle indicates the bioink with the suitable printing characteristics [59]. Reproduced from [58]. CC BY 3.0. (b) Nomogram for the mass flow rate and the maximum shear stress at the outlet of the printing needle as functions of the outlet diameter and pressure. Reproduced from [60]. CC BY 3.0.

After extrusion, the droplet forms a spherical shape before landing on the platform, where it spreads out upon collision. Due to the combined effects of inertial force, surface tension, and viscosity, the droplet retracts to form a hemisphere. Each droplet individually reaches the substrate surface, collectively forming a linear structure. A mathematical simulation proposed to analyse the landing process showed that a larger surface contact angle leads to a broader spreading radius, faster transformation, and thicker film formation. Conversely, higher droplet velocity causes wider spreading and potential separation into scattered shapes [69].

Direct printing onto objects with complex surfaces is common in the medical field. Arango and colleagues proposed a mathematical model to compensate for the distance from the nozzle to the non-flat substrate based on energy balance [70]. Triangular prisms and truncated cylinders were used to verify experimental and simulation results. Using high-density bioink and larger volume droplets helps prevent deviations from the intended target, though it reduces printing resolution and may cause rheological issues. This underscores the importance of optimising printing parameters, especially for curved surfaces. The summary of simulation used in inkjet printing has been listed in table 2.

3.2. ME

ME is another printing technique, which continuously deposits materials to create objects. The materials used in ME can include bioink or thermoplastic filaments. The morphology of the printed materials is important as it affects the printing resolution, surface quality, and structural integrity of the final product. Various printing parameters such as pressure, nozzle size, melting temperature, extrusion speed, and distance from nozzle to platform play significant roles in determining the quality of the final products. A comprehensive understanding of the ME process is essential for optimising strength, speed, and fidelity of the printed product. Table 3 provides a summary of simulation-based prediction methods pertinent to this process.

The bioink used in the ME is same as that used in inkjet printing. The primary difference between ME and inkjet printing is that the ME continuously deposits materials rather than creating droplet. Therefore, the preparation of bioink-based materials for ME can follow similar protocols to those used in inkjet printing. Another material used in the ME is the thermoplastic filament, which is manufactured to specific diameter specifications. The filaments used for ME are usually mixed with appropriate drug concentrations and exhibit adequate mechanical properties to ensure they can be loaded into the printer without issue. Filaments should possess the desired mechanical properties. Otherwise, they may suffer from various issues such as 1. breakage by feeding gears if the materials are too fragile; 2. undeliverability if the material is too soft; or 3. surface breakdown due to low toughness. Modelling printability and passive drug diffusion into the filaments is crucial for developing new filament materials. These filaments are heated above their melting point before being extruded from the nozzle, and as the filament transitions from a solid to a liquid phase, its rheological properties significantly impact the printing accuracy and sensitivity.

There are two main methods for drug loading in ME: hot-melt mixture and passive diffusion in tablet printing. The hot-melt mixture method requires careful consideration of the melting temperatures of both the filaments and the drugs, as excessively high temperatures can degrade or inactivate the drugs. The major limitation of the passive diffusion method is the maximum drug loading percentage that can be incorporated into the filaments. Previous reports

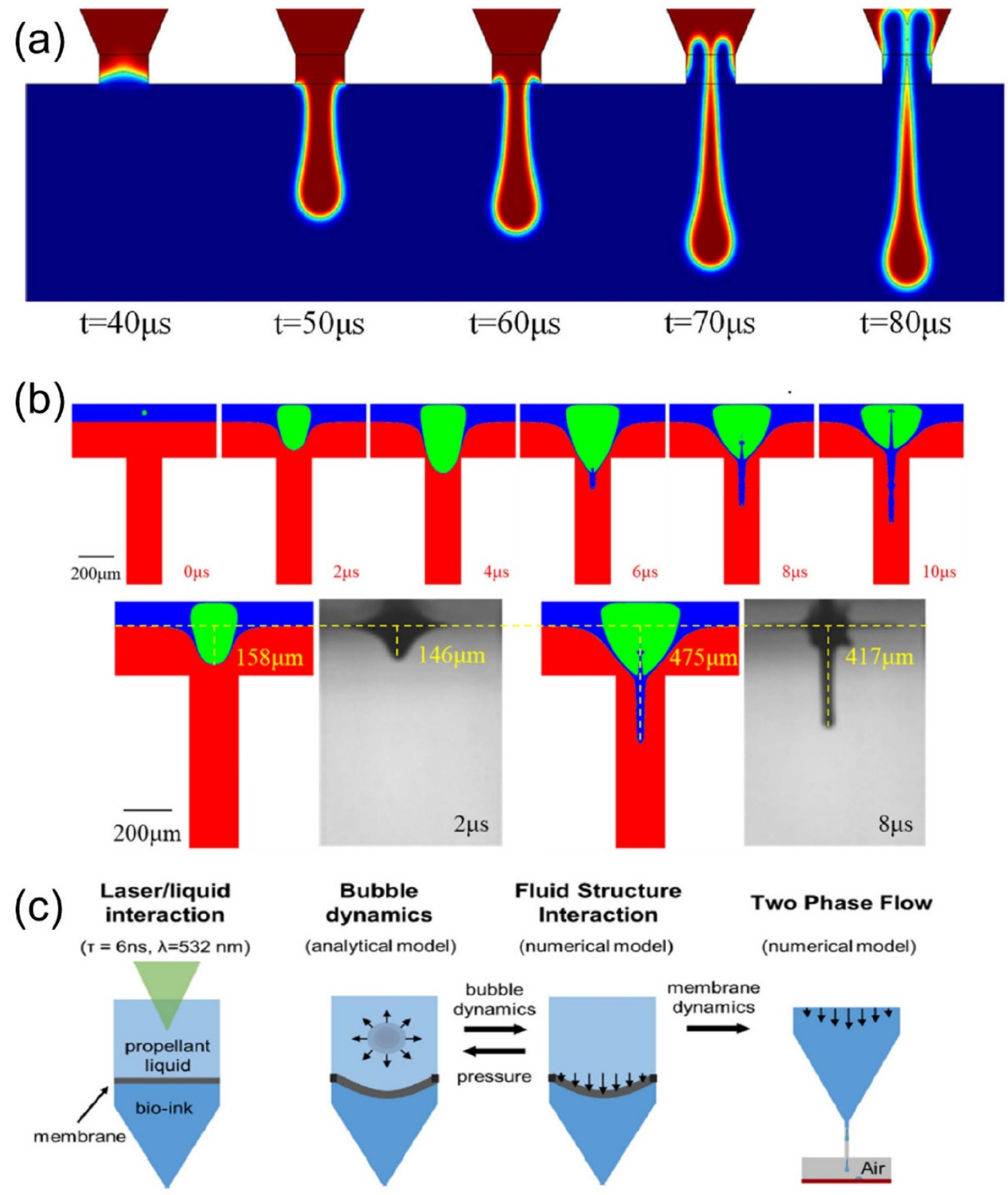

Figure 4. (a) Droplet formation process by hybrid drive. Reproduced from [36], with permission from springer nature. (b) simulation of jet flow compared with experiments. Reprinted from [67], with the permission of AIP Publishing. (c) schematic of the laser-based microdroplets printing process. Reprinted from [68], Copyright (2020), with permission from Elsevier.

indicate that drug-loaded percentage is less than 2% w/w using passive diffusion method [90, 91]. The printability of a new filament can be predicted by performing a bending test [72, 92]. A decision tree as shown in figure 5 illustrates the relationship between the printability of filaments and the maximum displacement and force in a bending test, enabling the prediction of filament printability based on these parameters.

After the materials have been designed to meet specific requirements and loaded into the printer, the next step is to analyse the effect of printing parameters on their physical and chemical performance post-extrusion. By simulating the dynamic deposition process, it is possible to understand the fluid dynamics of the filaments from nozzle to platform, optimise the shear forces associated with viscosity, and balance the cell viability or the quality of printed products with the printing parameters. The shape of the printing nozzle can significantly affect shear stress. For instance, a tapered conical nozzle generates lower shear stress compared with a conical nozzle (figure 6(a)). Besides, decreasing pressure and increasing nozzle diameter can significantly reduce shear force [73, 74]. Göhl *et al* assessed the printed shape in relation to printing parameters, predicting the effects of nozzle height, line width, and printing speed on the shape and viscoelastic stress

Table 2. Summary of inkjet printing simulation.

Type	Aims	Methods or software	References
Inkjet	Predict printability of droplet	Calculate *Re, We* and *Oh* numbers	[58]
Inkjet	Predict printability of droplet	Calculate Fromm's *Z* parameter via acquiring dynamic viscosity, surface and density of the ink	[61]
Direct ink writing	Design multi material inks to match rheological and mechanical properties	Calculate particle interaction potential via particle volume fraction/size/size distribution, and precursor concentration	[62]
Inkjet	Predict droplet formation	The volume-of-fluid method and continue surface force model	[64]
Inkjet	Effect of temperature on shear stress, pressure, velocity	COMSOL Multiphysics 5.4a	[65]
Inkjet	Model the flow behaviour and rate of bioink	Herschel–Bulkley model	[66]
Inkjet	Optimise a high-temperature microheater	COMSOL 5.1. A	[42]
Inkjet	Optimise printing parameters	Ansys Workbench18.2	[60]
Inkjet	Predict the flow state in the channel	N–S equation and linear time-varying system equivalent circuit model	[34]
Inkjet	Pressure field control and optimisation	N–S equation	[35]
Inkjet	Predict droplet formation	N–S equation	[36]
Inkjet	Predict droplet formation	N–S equation	[67]
Inkjet	Predict ink extrusion process	The rayleigh–plesset equation combined with COMSOL	[68]
Inkjet	Predict the printing process	Numerical calculation combined with finite element method	[69]
Inkjet	Predict drop distance of the ink in non-planar substrate	Numerical calculation	[70]
Direct ink writing	Predict non-Newtonian inks printing process	Lattice Boltzmann model	[37]
Direct ink writing	Predict equivalent stress of printed valve	ANSYS	[71]

Table 3. Summary of ME-based simulation.

Aims	Methods	References
Predict printability of filaments	3-point bending test and decision tree	[72]
Nozzle optimisation	Ansys Fluent and COMSOL Multiphysics 5.4a	[73, 74]
Predict printed filaments morphology	The volume of fluid and the interFoam solver	[75]
Control the behaviour of deposited lines at corners for moderate viscosity inks	Numerical calculation	[76]
Control the direction-dependent microstructures	Numerical calculation	[77]
Predict bioink flow behaviour in extrusion method	Finite element analysis	[78–80]
Predict filament deformation	Finite element analysis	[81]
Predict temperature evolution	Domain discretisation combines Fourier's equation	[82]
Predict temperature evolution and crystalisation	Domain discretisation combines Fourier's equation	[83]
Predict part distortion in materials extrusion of semi-crystalline polymers	COMSOL	[84]
Predict mechanical property of passive-diffused filaments	Finite element methods	[85]
Predict the fill density of final products	Numerical calculation	[86]
Predict mechanical properties of printed products	Fourier's equation and numerical calculation	[87]
Predict mechanical behaviour of material printing direction	Classical laminate theory	[88]
Predict drug release profile	Fick's law	[89]

distribution of extruded materials [93]. The extrusion velocity is the major parameter affecting morphology changes, while surface tension and shear viscosity have minimal influence (figure 6(b)) [75]. Achieving high printing resolution at corners or turns during the printing process is challenging due to factors like capillarity, nozzle retracing, and printer vibration [76]. A mathematical model was built to investigate the influence of three printing parameters, namely, printing direction, corner accuracy and fluid support on ink extrusion. The proposed model can predict the particle distribution and width difference between corner and centre, including these 3 factors, allowing for the controlled distribution of particle-loaded materials in ME [76, 77, 94].

For the fabrication of hollow structures, such as vascular regeneration, a coaxial ME method is particularly suitable (figure 6(c)). The shear stress within outflow channel significantly influences the distribution and survival rate of cells. The simulation

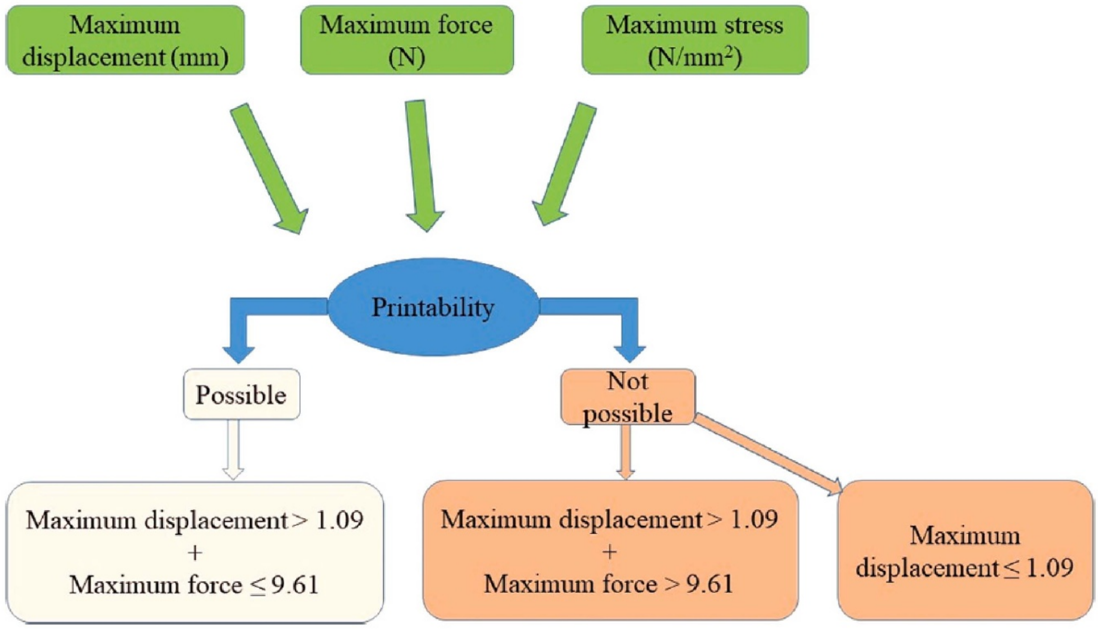

Figure 5. A decision tree shows the relationship between filament printability with maximum displacement and force in the 3-point bending test. Reprinted from [72], Copyright (2021), with permission from Elsevier.

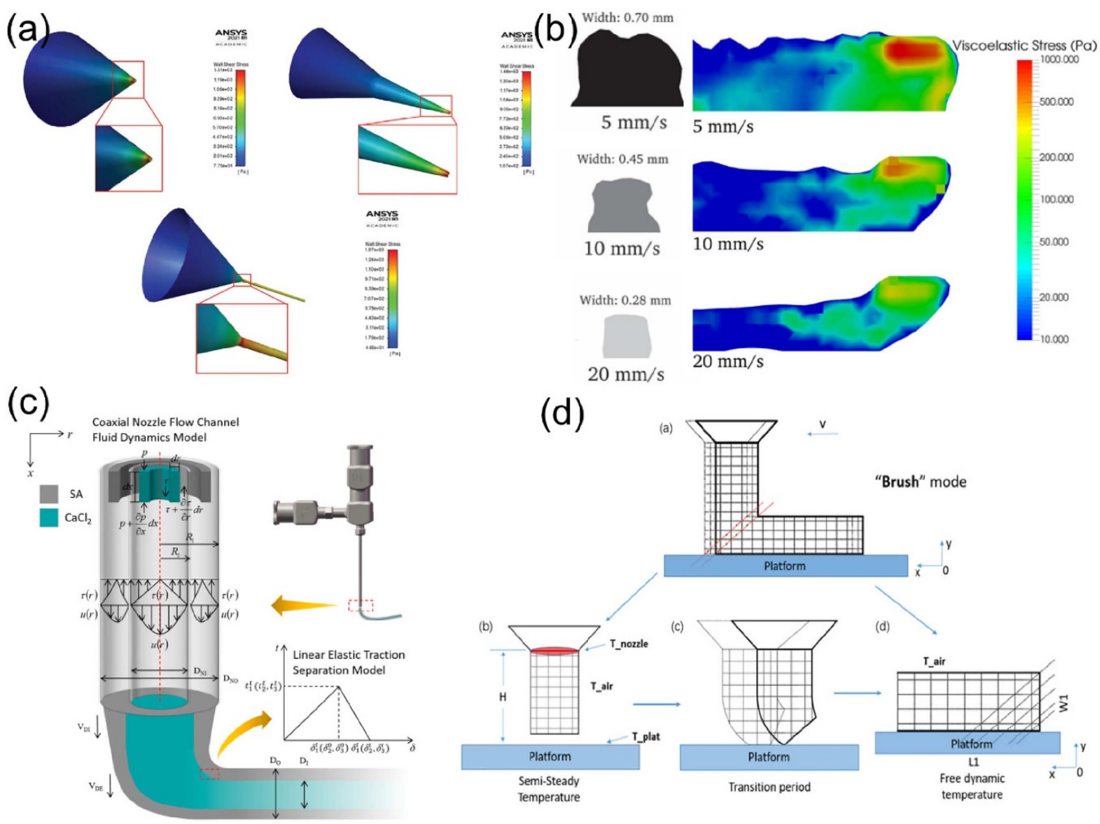

Figure 6. (a) Contour of wall shear stress for tapered conical, conical, and cylindrical nozzle. Reproduced from [73]. CC BY 4.0. (b) comparison of the cross-section area and distribution of viscoelastic stresses in lines printed with different printing speed. Reproduced from [75]. CC BY 4.0. (c) schematic diagram of the theoretical model for each stage of the coaxial jet process. Reproduced from [80]. CC BY 4.0. (d) meshed filaments of three periods in time sequence starting from semi-steady temperature to transition period and finally, free-dynamic temperature during ME printing process. Reprinted from [83], Copyright (2020), with permission from Elsevier.

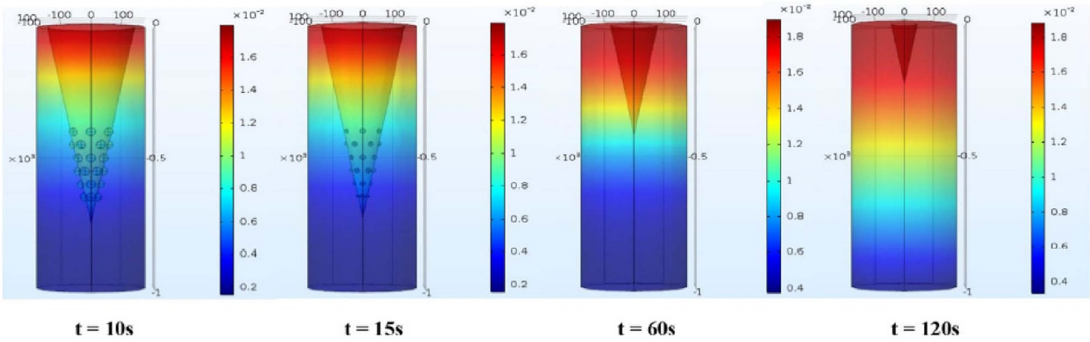

Figure 7. The microneedle dissolution process with time. Reprinted from [99], Copyright (2020), with permission from Elsevier.

model can assist in analysing flow behaviour and extrusion process, optimising the input pressure. To balance the mechanical property of fabricated vascular with printing materials, the formulation and printing parameters can be optimised to meet the requirements [78–81].

Thermal dynamic presents another challenge in 3D printing fabrication, inducing morphology changes, and limiting the prediction of printing accuracy. The proposed models used numerical finite-difference approaches to predict temperature changes (figure 6(d)) [82–84]. These models accounted for phase changes through latent heat resources and included critical parameters such as thermal convection coefficient, thermal conductivity, nozzle diameter, and printing speed. By modelling and optimising these parameters, spatial heat administration can be controlled and homogenised, preventing thermal and crystalisation damage to feedstocks during the development of new multi-printing drugs. Additionally, adjusting these critical factors can accelerate printing speed while maintaining product functionality, thereby enhancing the production process.

To further improve the precision and quality of the printing process, it is recommended to combine simulation with digital twin in-process monitoring or cyber physical system, enabling real-time adjustments of the targeted products. These systems can detect the edges of printed parts and compare them with the original CAD model to assess differences. By using the information gathered from these computational algorithms based, real-time data analysis, and control systems, users can ensure that the deposited filaments remain within the acceptable threshold distance [95, 96].

The various prediction in device design, materials properties and characteristics, and printing processing parameters have been systematically reviewed above. The following sections will focus on the simulation of the ME printed products and drug-release kinetics.

For printed products, mathematical simulation and FEM are frequently utilised to simulate mechanical properties [71, 85, 97]. In ME printing, fill density is critical in determining the mechanical property, porosity, and structure degradation of printed products. Fill density is particularly vital in biomedical applications, as porosity is a key parameter that affects the additive diffusion rate, subsequently influencing the growth and proliferation of cells seeded within the construct [86]. Void density between printed filaments is crucial for the quality of final printed products. Controlling the temperature could improve the sintering processing and reduce the void density [87], hence improving the mechanical properties of the final product. Besides, the mechanical property in the transverse direction is always weaker than the axial direction of printed products. Optimal selection of printing direction and parameters ensures that the final products meet the required mechanical properties [88]. Besides structural optimisation, composite interlocking scaffolds with different formulations can also facilitate load transfer and enhance interfacial integration. Soft materials, due to their larger deformation, are always unsuitable as solo materials for fabricating interlocking scaffolds. When the difference in Young's modulus between materials is less than an order of magnitude, the combination of soft and hard materials can result in the lowest von Mises stresses while maximising load transfer abilities [98].

In addition to structural optimisation, ME-based printing techniques are increasingly applied to pharmaceutical applications beyond scaffolds, such as drug delivery systems. One notable example is the fabrication of microneedles, which offer a minimally invasive and more convenient alternative to conventional needle injections. These dissolvable microneedles allow for controlled drug administration, where the diffusion process through the skin can be simulated to optimise release profiles and degradation behaviour (figure 7) [99]. Through the simulation, the drug release profile and degradation can be replicated *in silico* and controlled drug administration can be designed by changing the materials, geometry, and drug loading of microneedles or capsules [89, 99].

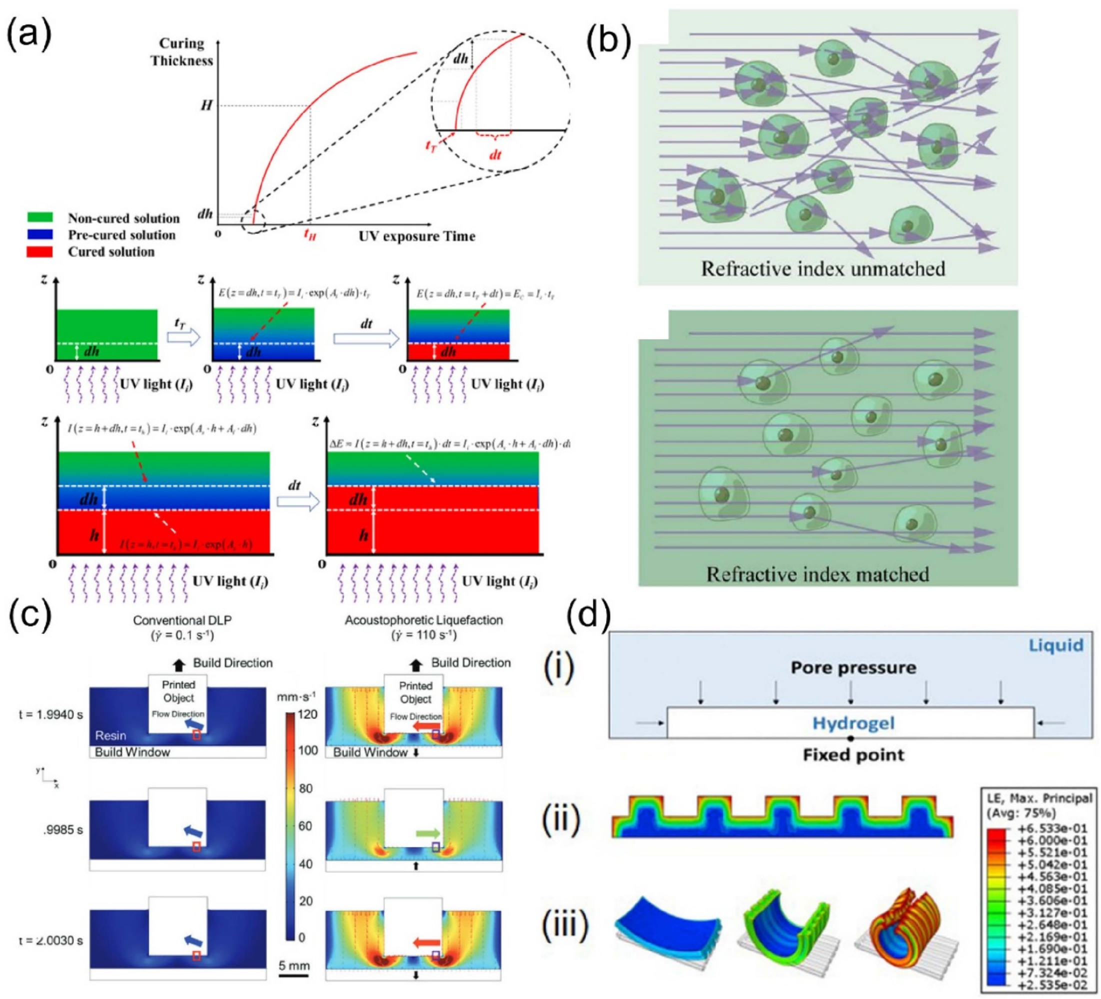

Figure 8. (a) Schematic diagram of a Jacobs working curve, the beginning of curing with an infinitesimal thickness dh, and curing with an infinitesimal thickness dh of a completely cured solid with a thickness of h. Reprinted from [100], Copyright (2020), with permission from Elsevier. (b) A schematic showing light propagation in a refractive index–unmatched and matched bioink. Reproduced from [57]. CC BY 4.0. (c) Simulation result of resin flow under acoustic. [101] John Wiley & Sons. © 2022 Facebook Technologies, LLC. Advanced Materials © 2022 Wiley-VCH GmbH. (d) Moisture swelling model used for the FEA simulations. Distribution of hydrostatic pressure and Shape morphing. The volume expansion started from the surface, and the deformation progressed due to the relative differences in the hydrogel interior in terms of the volume expansion rate. Reprinted from [102], Copyright (2020), with permission from Elsevier.

3.3. DLP

There are relatively fewer simulation studies on DLP printing than on inkjet and ME printing. In DLP printing, simulations primarily aim to predict the photo-crosslinking process, and the quality of printed products based on relevant printing parameters. These parameters include bioink properties, light intensity, light exposure time, print layer thickness, geometry, and physicochemical and mechanical properties.

In DLP printing, the light exposure plays a critical role in determining the photopolymerisation process. The photocuring process was normally divided into 3 steps: (1) pre-curing; (2) completion of curing; and (3) preparation for the next curing cycle. The energy absorption happened in the first two stages following Beer's law [54]. By determining values such as solid absorbance, liquid absorbance, and gelation time, the curing time of each layer can be predicted using Jacobs' curve. This curve illustrates the relationship between absorbed UV light time and the thickness of solidified photocurable material (figure 8(a)) [100].

Functional gradient structures can be achieved using the DLP method by varying the greyscale light patterns. A radiative transfer coupled chemical reaction—diffusion model was proposed to simulate how different greyscale light patterns affect the quality of printed layers. Light exposure triggers photoinitiators to break down into free radicals, which then initiate localised polymerisation. As a result, printed shapes, especially at pattern boundaries, exhibit significant variations with varying greyscale light intensities [103]. Light scattering can also occur when particles or high-density cells are loaded in bioink [104]. In cell-loaded bioink, light scattering primarily arises from two sources: the refractive index mismatch between the cytoplasm and the surrounding biomaterial, and subcellular components such as the nucleus and organelles. To mitigate the former effect, which can be effectively reduced by incorporating iodixanol, a biocompatible supplement (figure 8(b)). A Monte Carlo-based model was utilised to simulate the propagation of light, demonstrating a 10-fold reduction in light scattering [57]. When the viscosity of bioink is high, removing partially solidified bioinks from the walls of printed products becomes challenging. These residual bioinks can cause overcuring during subsequent curing processes, thereby reducing the final resolution of printed products. Moreover, high viscosity bioinks can lead to significant adhesive forces, potentially resulting in product failure. An acoustic approach was proposed to enhance the bioink flow and prevent overcuring. Numerical simulations can predict the velocity distribution of bioink in the system under varying frequencies of acoustic agitation. Optimising the frequency through simulation aims to maximise ink flow rates and enhance replenishment of the build area (figure 8(c)) [101].

DLP-printed products exhibit anisotropic properties along the z-axis due to the layer-by-layer printing process. When printed hydrogels are immersed in water, their designed structure leads to differential swelling and expansion rates, resulting in internal forces that cause directional deformation of the printed structure (figure 8(d)) [102, 105]. In another study, varying tablet thickness and drug loading can adjust the drug release profile [106].

4. Discussion and future perspectives

Numerical simulation in 3D bioprinting remains in its early stages. Despite this, it has become an essential theoretical tool for understanding the physical, chemical, and biological aspects of the printing process. Simulation enables the prediction, optimisation, and control of key parameters influencing print quality, structural fidelity, and cell viability. Many of the simulation studies discussed in this review have already been validated experimentally, confirming the accuracy of their numerical predictions, and these experimental results are typically reported together with the corresponding modelling frameworks in the original studies [36, 68, 83].

In addition, the simulation frameworks described here are largely mechanism-based and derived from fundamental physical principles. Therefore, these models are capable of being applied to a wide range of bioink/printing systems rather than being limited to specific materials/device [36, 82–84, 99]. Furthermore, many simulation and experimental studies have systematically examined the influence of printing parameters, including extrusion pressure, nozzle diameter, temperature, and cell density on the quality and structural stability of bioprinted constructs [57, 67, 73, 75]. These same parameters are commonly incorporated into simulation models to analyse flow behaviour, droplet formation, and deposition stability. By integrating multiple theoretical domains, including fluid dynamics, continuum mechanics, and light–matter interaction, simulation provides a framework for bridging engineering theory with experimentally observed printing performance in biological systems.

4.1. Simulation classification

Although AM technologies are commonly classified according to the ISO/ASTM 52 900 framework, such a classification mainly reflects differences in fabrication mechanisms rather than modelling strategies. From a simulation perspective, alternative classfication approaches could be considered, for example grouping models according to the underlying physical principles, including fluid dynamics, mechanical deformation, heat transfer, or photopolymerisation kinetics. These simulation-oriented perspectives may provide additional insights into how different modelling strategies can be transferred across printing platforms.

However, establishing a comprehensive classification based on simulation methodologies would require a systematic comparison of modelling frameworks across multiple application domains and physical processes. Therefore, this work follows the widely accepted ISO/ASTM classification while focusing on how similar theoretical models are applied and adapted within different bioprinting technologies. This approach allows the discussion to remain closely connected to practical printing processes while highlighting the shared theoretical foundations that support simulation studies in inkjet printing, ME, and DLP.

Nonetheless, we have introduced a secondary perspective by classifying simulation approaches according to the state of the simulated objects, including liquid, solid, particle, and multiphase systems in table 1.

4.2. Limitations and future directions

Many current simulations still rely on simplified physical assumptions, particularly those treating bioinks as Newtonian fluids [35, 60, 64]. While this may capture basic flow dynamics, it does not represent the viscoelastic, shear-thinning, or yield-stress behaviours characteristic of most cell-laden bioinks. These rheological complexities are critical, as they strongly influence droplet formation, extrusion stability, and post-deposition shape fidelity. As non-Newtonian fluid behaviour is a fundamental aspect of many 3D printing and bioprinting processes, particularly in describing bioink flow, droplet formation, and extrusion stability, the extensive literature on non-Newtonian fluid modelling in related fields, such as microelectronics and dispensing processes, provides valuable insights into fluid

Table 4. Limitations and future directions of theory-based simulation models in biomedical and pharmaceutical 3D bioprinting.

Aspect	Current limitation	Potential solutions	Expected impact	References
Bioink rheology representation	• Most models assume Newtonian behaviour • Ignore viscoelastic, shear-thinning, and yield-stress effects • Limited temperature-dependent rheology data	• Introduce Herschel–Bulkley or Carreau–Yasuda models • Incorporate temperature- and shear-dependent viscosity • Couple experimental rheological data with simulation	• Improved accuracy of droplet/extrusion prediction • Better bioink design and process optimisation	[110, 111]
Lack of coupled physics	• Separate treatment of mechanical, thermal, and optical processes • Simplified boundary conditions • Incomplete description of polymerisation kinetics	• Develop multi-physics models linking flow, heat, and curing • Use coupling modules • Include photo-polymerisation and diffusion parameters	• Realistic prediction of curing and mechanical integrity • Enhanced print resolution and structure stability	[54, 71, 100, 112]
Limited biological relevance	• Treats printed constructs as inert materials • Ignores cell–material interactions • No prediction of viability or biological function	• Integrate cell mechanics and biological transport • Couple mechanical stress with nutrient and oxygen diffusion • Validate using *in vitro* viability data	• Predict cell behaviour post-printing • Improve design of cell-laden bioinks	[58]
Simplified geometrical modelling	• Uses 2D or simplified nozzle/lattice geometries • Does not capture scaffold porosity or anisotropy • High computational demand for complex structures	• Apply adaptive meshing and topology optimisation • Use voxel-based or lattice-resolved models	• Higher geometric fidelity • More accurate structure–property prediction	[114, 115]
High computational cost	• Fine meshing and multi-physics coupling increase run time • Not suitable for real-time or iterative optimisation	• Combine physics-based models with machine learning • Develop surrogate models for fast prediction	• Reduce computational cost • Real-time process feedback	[116, 117]
Lack of validation and standardisation	• No universal benchmarks or test datasets • Inconsistent experimental conditions • Rare quantitative model verification	• Develop community benchmarks and open datasets • Integrate simulation with *in situ* imaging or sensors • Encourage data sharing through repositories	• Improved reproducibility • Stronger link between simulation and experiment	[132]

handling, spreading, and process control [107–109]. Incorporating more advanced constitutive models such as Herschel–Bulkley or Carreau–Yasuda formulations would therefore improve predictive accuracy, especially under variable shear and temperature conditions [110, 111].

Another limitation lies in the lack of coupled physics modelling. In many cases, mechanical deformation, heat transfer, and optical curing are simulated independently, resulting in incomplete or inconsistent process descriptions [54, 71, 100, 112]. For instance, in extrusion printing using a support bath, multiple physical processes occur simultaneously and influence the final printing outcome. In addition to the flow behaviour of the bioink during extrusion, the rheological properties of the supporting medium, such as yield stress, viscoelasticity, and shear recovery, must also be considered, as they determine the ability of the bath to stabilise deposited filaments. The interaction between the extruded filament and the surrounding support material further affects filament deformation, diffusion of solvents or biomolecules, and the maintenance of printing resolution [113]. Also, in DLP-based systems, polymerisation kinetics depend not only on light exposure but also on local temperature and diffusion conditions [54, 100]. Integrating these parameters through multi-physics models would enable a more realistic representation of curing dynamics and improve the reliability of predictions related to crosslink density and mechanical integrity.

Biological relevance remains an even greater challenge. Despite the term bioprinting, most existing models treat the printed medium as an inert material, neglecting the effects of embedded cells, nutrient diffusion, and post-printing biological responses [58]. As a result, few models can directly predict outcomes such as cell viability, proliferation, or differentiation. Future developments should incorporate cell mechanics and biochemical transport into mechanical simulations, allowing prediction of how local stress or shear distributions affect biological performance.

Furthermore, the simplification of geometrical domains continues to restrict the application of models to realistic tissue constructs. Simplified nozzle or lattice geometries are often used to reduce computational complexity, but they fail to capture the structural anisotropy and porosity of actual scaffolds [75, 99]. Advanced computational methods such as adaptive meshing, topology optimisation, and voxel-based finite element methods can help address these limitations by balancing accuracy with efficiency [114, 115].

The high computational cost of fine-meshed and multi-physics simulations also hinders their use in real-time or iterative design environments. Hybrid strategies that combine classical physics-based solvers with data-driven or ML approaches can significantly reduce computation time [116, 117]. These surrogate models can be trained using precomputed datasets to provide near-instant predictions of printability, mechanical strength, or drug diffusion, making them promising tools for digital twin integration.

Beyond physics-based modelling, ML has emerged as a complementary approach for analysing, predicting, and optimising printing outcomes. ML algorithms such as support vector machines (SVM), random forests (RF), and artificial neural networks (ANN) have been applied to predict printability, detect errors, estimate drug release rates, and assess mechanical performance and quality control [106, 118–130]. Despite their promise, ML approaches face several challenges, including limited datasets, data imbalance, and the risk of overfitting. Small-dataset ML techniques and hybrid data augmentation strategies have been introduced to overcome these issues [119, 131]. Integrating traditional physics-based models with ML can reduce the dependence on large datasets, accelerate prediction, and improve generalisation across different printing systems [116, 117]. Therefore, there is an urgent need for standardised validation and benchmarking of simulation results. At present, there is no universal framework for comparing model predictions with experimental or *in situ* monitoring data. Establishing such standards would allow consistent evaluation of model performance and strengthen the link between simulation and experimental practice [132]. The integration of digital twin systems and high-speed imaging could enable continuous feedback, closing the loop between simulation, monitoring, and process control.

Together, these considerations highlight that while significant progress has been made in applying theory-based models to bioprinting, their true translational value depends on improved physical realism, biological integration, and computational efficiency. As outlined in table 4, overcoming these challenges will be key to advancing simulation from a predictive tool to a decision-making component in bioprinting research and manufacturing.

5. Conclusion

We have reviewed recent advancements in theory-based simulations that contribute to optimising the 3D printing process for biomedical and pharmaceutical applications. These processes encompass formulating feedstock composition, morphology, and size distribution; selecting printing and binding techniques; optimising printing parameters; and performing post-printing profiling. The simulation methods can reduce experimental costs and facilitate the development of new materials, printing methods, and functional structural designs. We believe that theory-based numerical simulation holds significant potential for refining the 3D printing process and improving the quality of printed products.

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Data availability statement

No new data were created or analysed in this study.

Conflict of interest

The authors declare no conflict of interest.

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